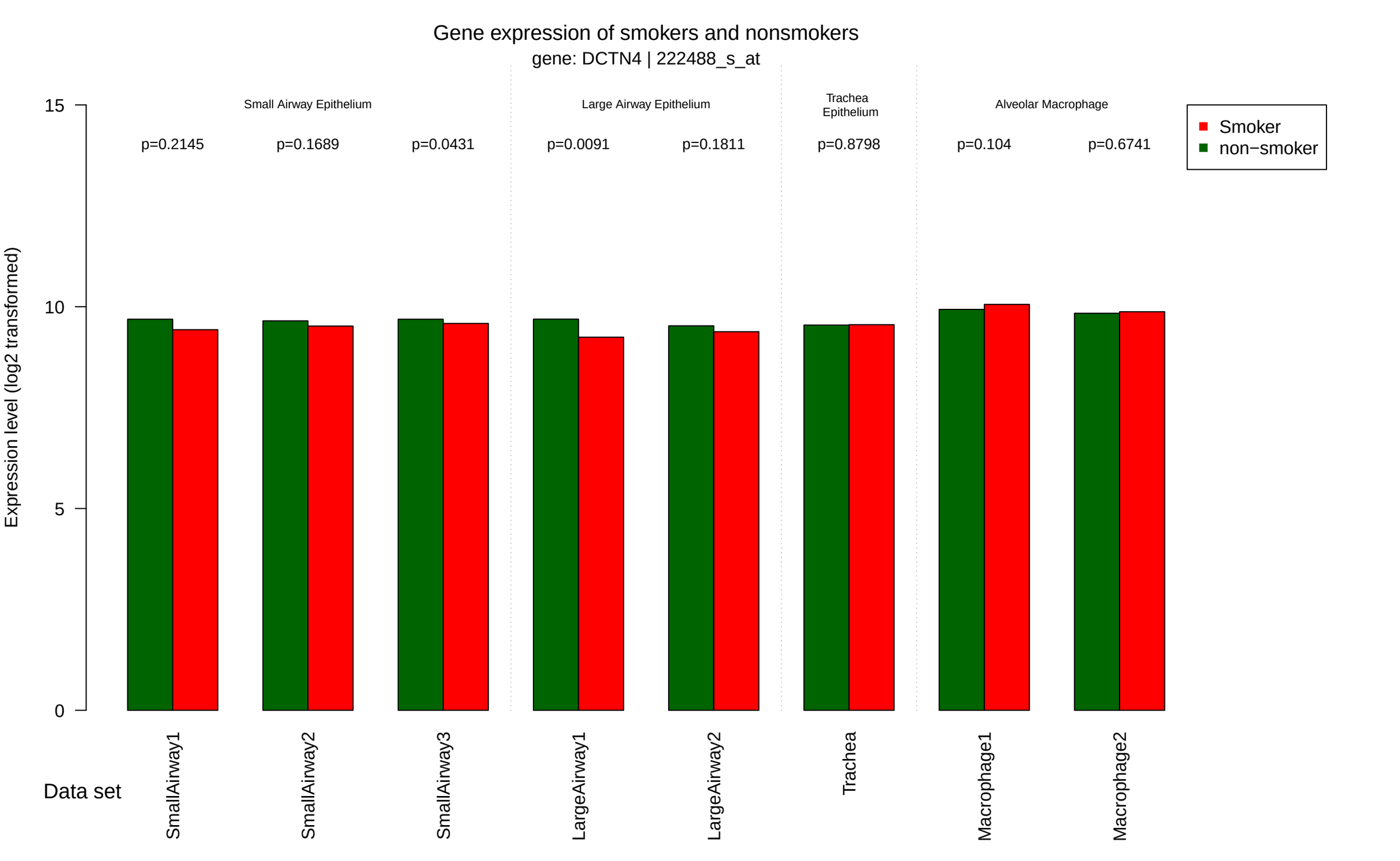
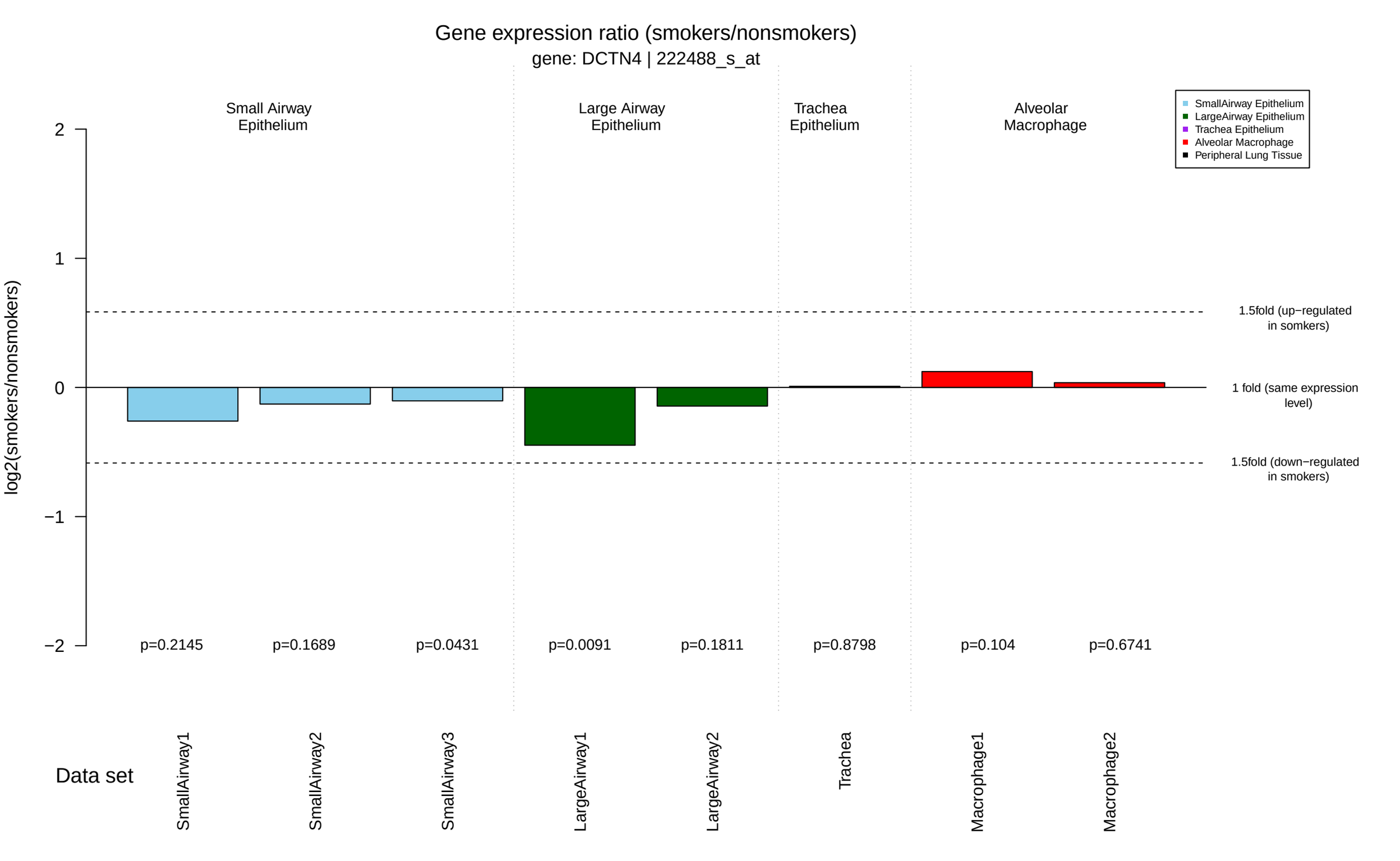
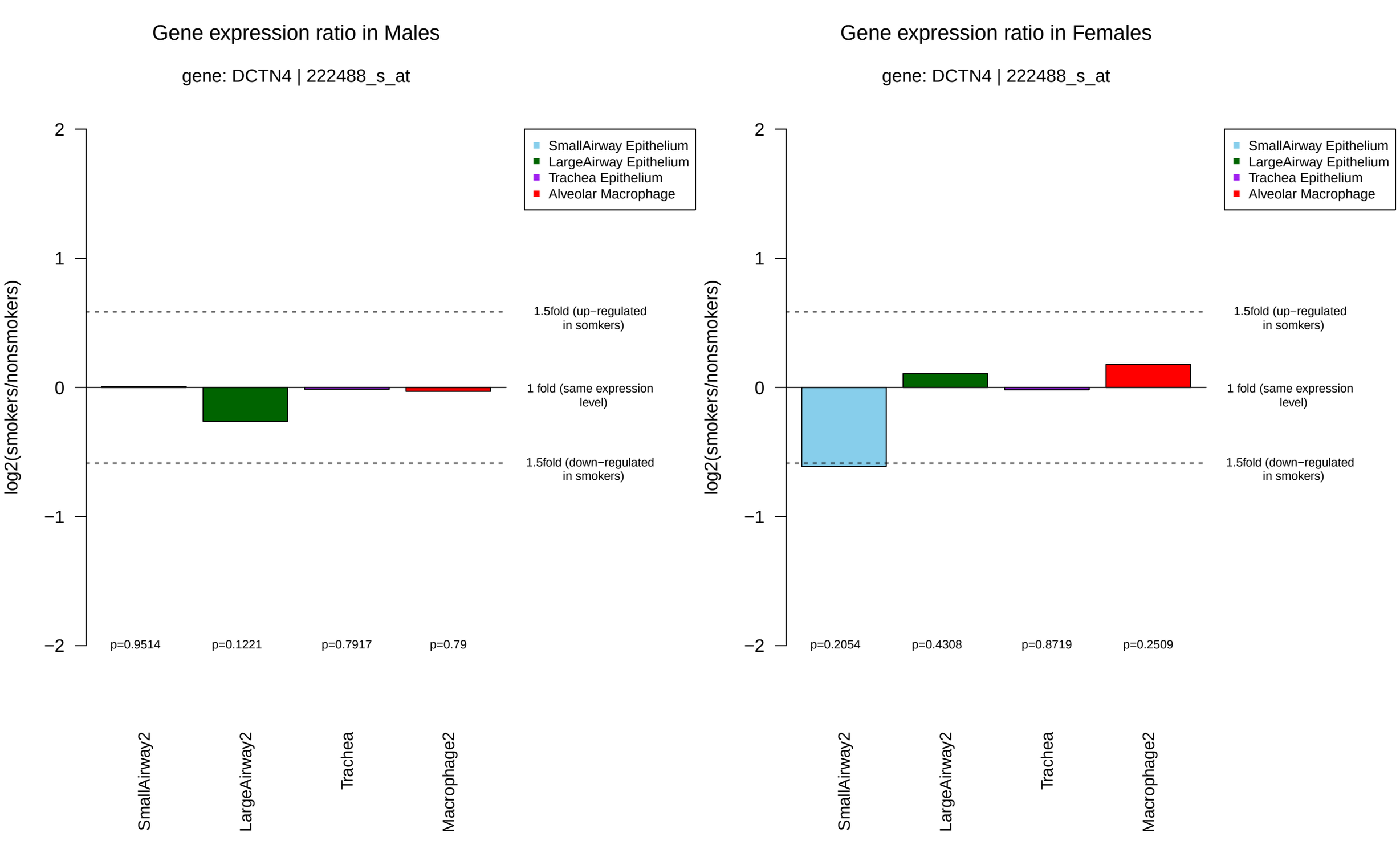
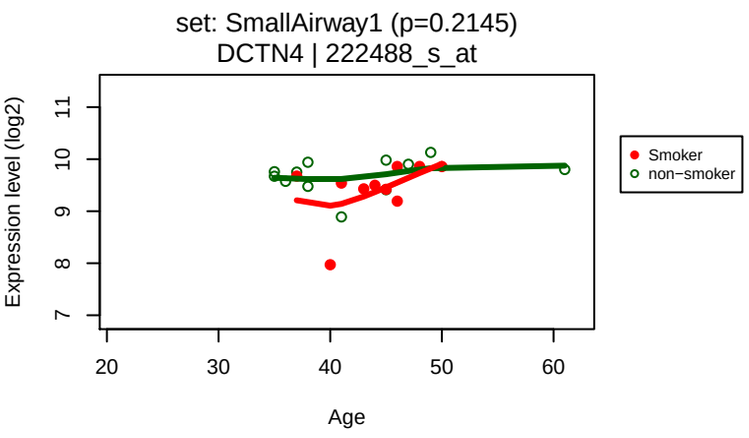
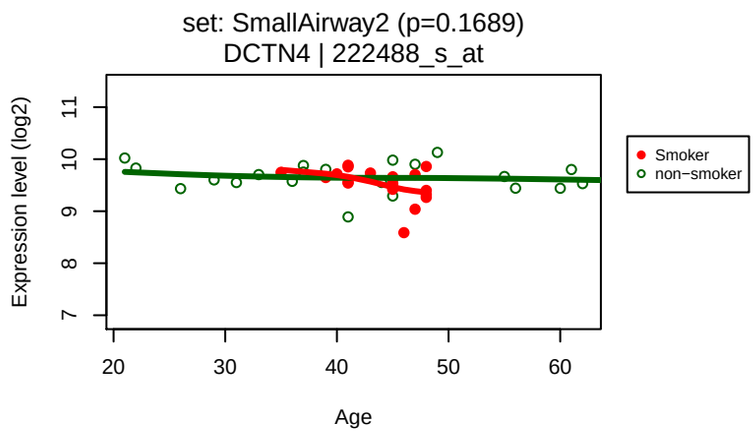
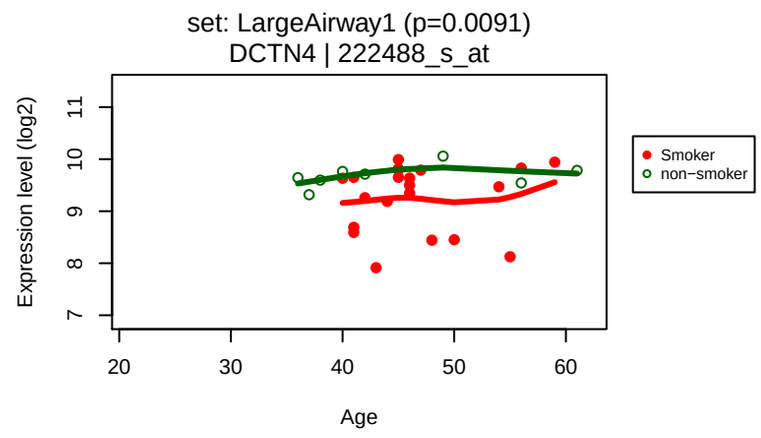

Gene expression of smokers and nonsmokers
gene: DCTN4 | 222488_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2145
p=0.1689
p=0.0431
p=0.0091
p=0.1811
p=0.8798
p=0.104
p=0.6741
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

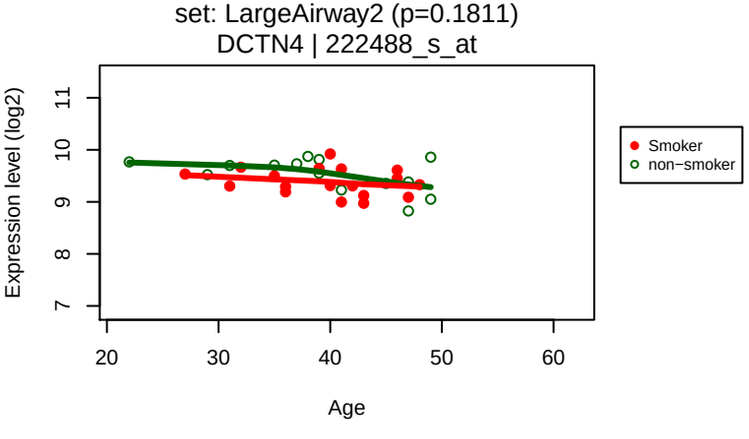

Gene expression ratio (smokers/nonsmokers)
gene: DCTN4 | 222488_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2145
p=0.1689
p=0.0431
p=0.0091
p=0.1811
p=0.8798
p=0.104
p=0.6741
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

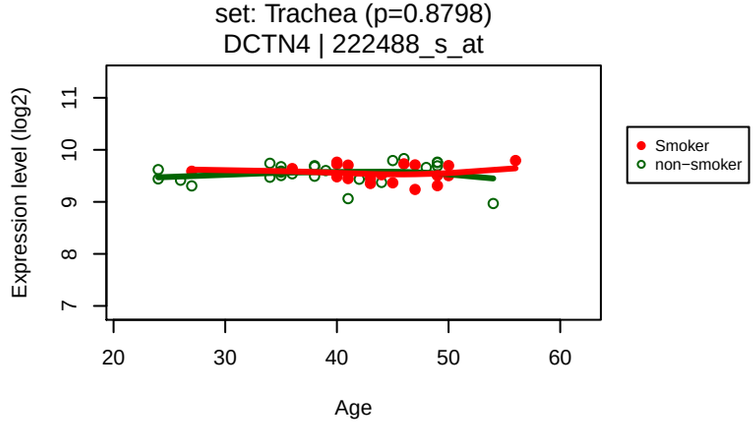

Gene expression ratio in Males
gene: DCTN4 | 222488_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9514
p=0.1221
p=0.7917
p=0.79
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DCTN4 | 222488_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2054
p=0.4308
p=0.8719
p=0.2509
SmallAirway2
LargeAirway2
Trachea
Macrophage2

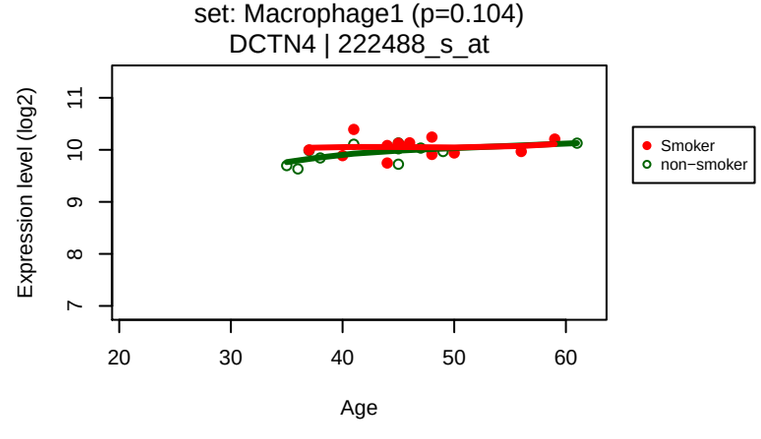

set: SmallAirway1 (p=0.2145)
DCTN4 | 222488_s_at
Expression level (log2)
Age
Smoker
non−smoker

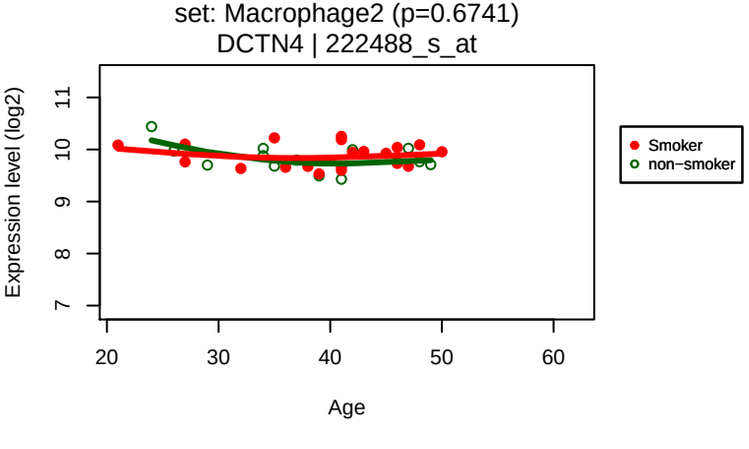

set: SmallAirway2 (p=0.1689)
DCTN4 | 222488_s_at
Expression level (log2)
Age
Smoker
non−smoker

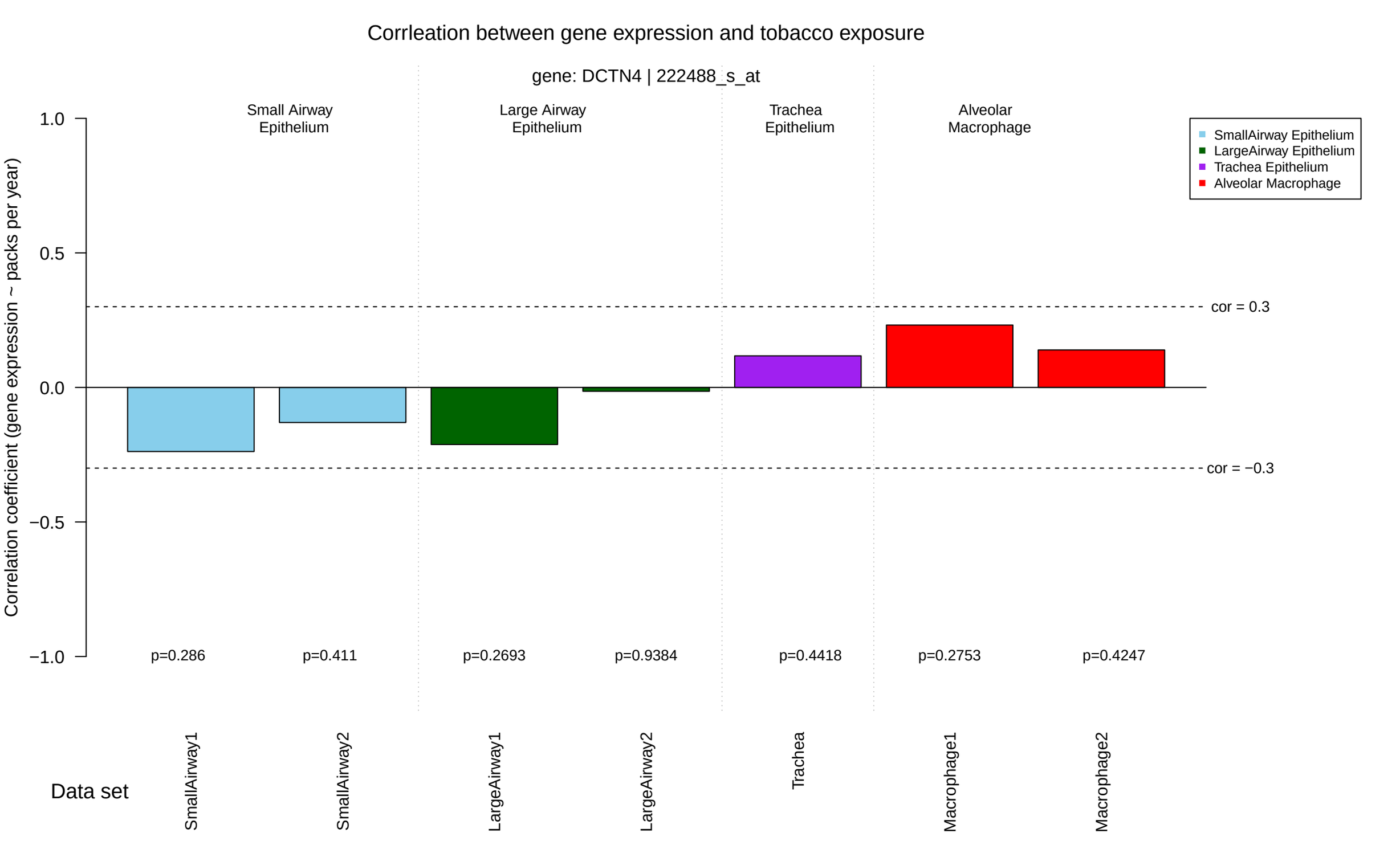

set: LargeAirway1 (p=0.0091)
DCTN4 | 222488_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1811)
DCTN4 | 222488_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8798)
DCTN4 | 222488_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.104)
DCTN4 | 222488_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6741)
DCTN4 | 222488_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DCTN4 | 222488_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.286
p=0.411
p=0.2693
p=0.9384
p=0.4418
p=0.2753
p=0.4247
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2